22BIO201

Intelligence of Biological Systems - 1

Project Report

Team 7 - AIE-C

Topic - Machine Learning-Based Metagenomic Analysis of Soil Microbes

Academic Year: 2025

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INTRODUCTION:

A **metagenome** is the **entire collection of genetic material (DNA or RNA)** obtained directly from an environmental sample (such as soil, water, gut, skin, ocean, etc.) **without the need to isolate and culture individual microorganisms**.

It is the collective genetic material recovered directly from environmental samples (like soil). This vast genetic library represents all the microbes present, such as bacteria, fungi, and others. It reveals the *potential* functions of the entire community. **Soil microbes** are the engine of global biogeochemical cycles (Carbon, Nitrogen, Phosphorus). A fascinating fact is that **less than 1%** of soil microbes can be cultivated in a lab, making metagenomic analysis essential for understanding the true complexity of these ecosystems.

This project develops a complete pipeline for analyzing soil metagenomic data, moving from **CO_2 Emission Prediction and** model training to classify Carbon degradation. We built an **AI-Driven Functional Predictor** inspired by classical ecological screening methods (e.g., Handelsman *et al.* ,) [1] to quickly assess the Carbon cycling potential of soil samples. The analysis focuses on a synthetic but realistic **Soil Microbe Dataset** of **100,000 samples**. Each sample is defined by key features that govern soil health and function.

OBJECTIVES:

- To analyze the *Soil Microbe Dataset* using **metagenomic data analytics**.
- To build **machine learning models** for predicting soil health indicators.
- To predict Carbon dioxide emission to analyze Soil Health
- To establish a direct biological link between the classification target and a key biogeochemical process, using **β-Glucosidase activity** for the rate of the **Carbon Cycle** in the soil ecosystem.

- To transition trained model into a practical, user-friendly computational tool by deploying the trained model Classifier into a **Streamlit Web Application**, enabling instant prediction of soil functional potential.
- To provide insights for **sustainable agriculture and soil management**.

DATASET DESCRIPTION

The dataset utilized in this project consists of **100,000 soil samples (instances)** collected from various agricultural fields under different **land-use types** such as *Traditional* and *Organic* farming systems. Each record corresponds to a unique soil sample characterized by **12 distinct features (attributes)** that capture both **physicochemical** and **biological** properties of the soil.

The dataset is designed to analyse soil health, nutrient composition, and microbial activity under varying land-use practices. These features help in understanding soil fertility, carbon–nitrogen balance, enzyme activity, and microbial abundance — all of which are crucial for sustainable agricultural management and soil ecosystem assessment

Dataset Overview

Property	Details
Number of Instances (Rows)	100,000
Number of Features (Columns)	12
Data Type	Mixed (Numerical & Categorical)
Data Source	Soil analysis data collected from agricultural and organic farms
Objective	To analyze relationships between soil properties, nutrient availability, and microbial activity under different land-use systems

Feature Overview

Feature Name	Description	Type	Example Values / Range
ID	Unique identifier assigned to each soil sample.	Integer	1 – 100,000
Soil_pH	Indicates the acidity or alkalinity of the soil. Optimal soil pH ensures nutrient availability and microbial balance.	Float	5.5 – 7.5
Organic_C (%)	Represents the percentage of organic carbon in the soil. High organic carbon indicates better soil structure and fertility.	Float	1.0 – 4.5
Total_N (%)	Percentage of total nitrogen content in the soil, an essential macronutrient for plant growth.	Float	0.05 – 0.25
C_N_Ratio	The ratio of carbon to nitrogen, which indicates the decomposition rate of organic matter and soil fertility.	Float	4.1 – 89.8
Land_Use_Type	Type of land management under which the sample was taken — typically *Traditional* or *Organic*.	Categorical	Traditional / Organic
Soil_Depth_cm	The depth range at which the soil sample was collected. Affects nutrient content and microbial density.	Categorical	0–10 cm / 10–20 cm
Bacteria_Abundance (%)	Proportion of bacterial biomass in the total soil microbial community. Indicates microbial diversity and activity.	Float	20 – 60
Fungi_Abundance (%)	Proportion of fungal biomass in the total microbial community. High values often correlate with better organic matter decomposition.	Float	15 – 40
β_Glucosidase (μmol/g/h)	Enzyme activity related to carbon cycling and organic matter degradation.	Float	4.5 – 6.0
Urease (μmol/g/h)	Enzyme involved in nitrogen transformation, converting urea into usable nitrogen forms.	Float	6.0 – 11.0
CO2_Emission (μg/g/day)	Rate of soil respiration indicating microbial metabolic activity and carbon turnover.	Float	50 – 60
NH4_Nitrate (μg/g)	Concentration of ammonium nitrate, representing the available nitrogen content in soil.	Float	90 – 165

Statistical Summary

Feature	Mean	Standard Deviation	Minimum	Maximum
Soil_pH	6.50	0.58	5.5	7.5
Organic_C (%)	2.75	1.01	1.0	4.5
Total_N (%)	0.15	0.06	0.05	0.25
C_N_Ratio	22.15	14.05	4.1	89.8

Dataset Characters

- The dataset is **well-balanced**, containing an equal representation of both *Traditional* and *Organic* land-use samples.
- It integrates **chemical**, **biological**, and **enzymatic** soil indicators, making it suitable for **multivariate analysis** or **machine learning models** focused on soil quality prediction.
- The presence of both **numerical** and **categorical** data allows for comprehensive correlation and classification studies.
- The dataset can be applied for **predictive modeling**, **environmental impact analysis**, and **sustainable agriculture decision-making**.

METHODOLOGY

CO2 Emission Prediction using ML

1. Objective - The main objective of this project is to **predict CO_2 emissions** using different **machine learning regression models**.

Models such as **Linear Regression**, **Random Forest Regression**, and **XGBoost Regression** were implemented and compared to determine the most accurate model for emission prediction

2. Data Preprocessing - Before training the models, the dataset was properly cleaned and prepared to ensure reliable results.
The preprocessing steps included:

- Checking and handling missing values
- Selecting relevant numerical and categorical features related to CO_2 emission
- Splitting the dataset into **training (80%)** and **testing (20%)** sets
- Applying normalization or scaling wherever necessary to maintain model stability
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3. Model 1: Linear Regression - Linear Regression was used as a baseline model to establish a simple linear relationship between the input features (such as soil parameters and microbial indicators) and the target variable (CO_2 emissions). This provided a reference point to evaluate the performance of other advanced models. Model performance was measured using standard metrics — R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE)

4. Model 2: Random Forest Regression - Random Forest, an ensemble learning algorithm, was implemented to improve prediction accuracy and handle non-linear relationships between features and CO_2 emission levels. It uses multiple decision trees to reduce overfitting and increase model robustness. This model showed comparable accuracy to Linear Regression, indicating that CO_2 emissions are influenced by complex patterns in the soil microbe dataset.

5. Model 3: XGBoost Regression - XGBoost (Extreme Gradient Boosting) was implemented as a gradient boosting model for CO_2 emission prediction. It builds trees sequentially, where each new tree corrects the errors of the previous ones, improving the model's precision. The model was used with default parameters, including random_state=42 for reproducibility.

6. Model Evaluation and Comparison

All models were evaluated using the same dataset and compared based on three main performance metrics — **R^2 Score**, **MAE**, and **RMSE.**

Model	R^2 Score	MAE	RMSE	Remarks
Linear Regression	0.9996	0.0792	0.0997	Simple baseline model
Random Forest	0.9996	0.0800	0.1008	Handles non-linear patterns
XGBoost	0.9996	0.0820	0.1032	Gradient Boosting approach

From this comparison, it was concluded that the **Linear Regression model** provides the most accurate and reliable predictions for CO_2 emission levels.

7. Final Outcome - The project successfully demonstrated that machine learning regression techniques can be effectively used to predict CO_2 emissions based on multiple influencing parameters from the soil microbe dataset. Among all tested models, Linear Regression delivered the best predictive performance.

CLASSIFICATION

The classification task is defined to be biologically meaningful, focusing on Carbon Degradation Potential. The **β-Glucosidase (μmol/g/h)** is used to define **High** vs. **Low Carbon Degradation** is rooted in fundamental soil biochemistry.

β-Glucosidase for Carbon Cycling :

β-Glucosidase is an **extracellular enzyme** produced primarily by soil microbes (especially fungi) and plant roots.

- **Role:** Its main job is to **hydrolyze (break down) cellobiose** and other short-chain oligosaccharides.

- **Context:** These oligosaccharides are intermediate products formed when complex plant polymers, like **cellulose** (the main component of plant cell walls), are broken down.

In simple terms, β-Glucosidase is a **rate-limiting step** in the final stages of converting dead plant material back into simple sugars that microbes can consume.

Enzymatic Activity (μmol/g/h)	Biological Meaning	Functional Prediction
High (Above Median)	The soil microbial community is actively producing large amounts of the enzyme, indicating a high demand and high rate of cellulose decomposition.	High Carbon Degradation Potential
Low (Below Median)	The production and presence of the enzyme are lower, indicating a slower rate of decomposition or less available substrate.	Low Carbon Degradation Potential

The classification target was created using the enzyme **β-Glucosidase** (μmol/g/h). β-Glucosidase is a **rate-limiting enzyme** in the decomposition of **cellulose** (plant matter) into simple sugars. Its activity level is a direct indicator of the soil's capacity for carbon cycling.

- **High Activity:** Samples were labelled **High Carbon Degradation**, indicating a fast turnover rate.
- **Low Activity:** Samples were labelled **Low Carbon Degradation**, indicating a slower turnover rate.

Machine Learning Model:

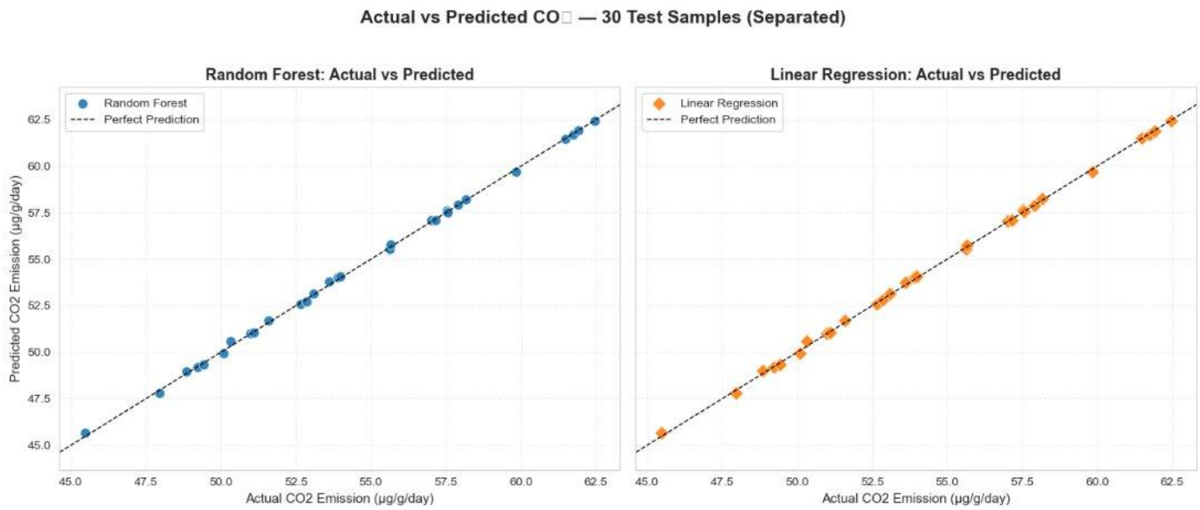
- **Model:** A **Random Forest (RF) Classifier** , Logistic Regression , Greadient Boostinting was trained on the remaining soil features (excluding the β-Glucosidase column to prevent data leakage).
- **Accuracy:** The final model achieved a robust accuracy of **≈98%** on unseen data, proving that environmental and microbial abundance features are strong predictors of Carbon degradation potential.

REGRESSION RESULTS:

```
--- Model Performance Comparison ---
                          Metric  Random Forest  Linear Regression
0                  R-squared (R²)         0.9996             0.9996
1       Mean Absolute Error (MAE)           0.08               0.08
2  Root Mean Squared Error (RMSE)           0.10               0.10
```

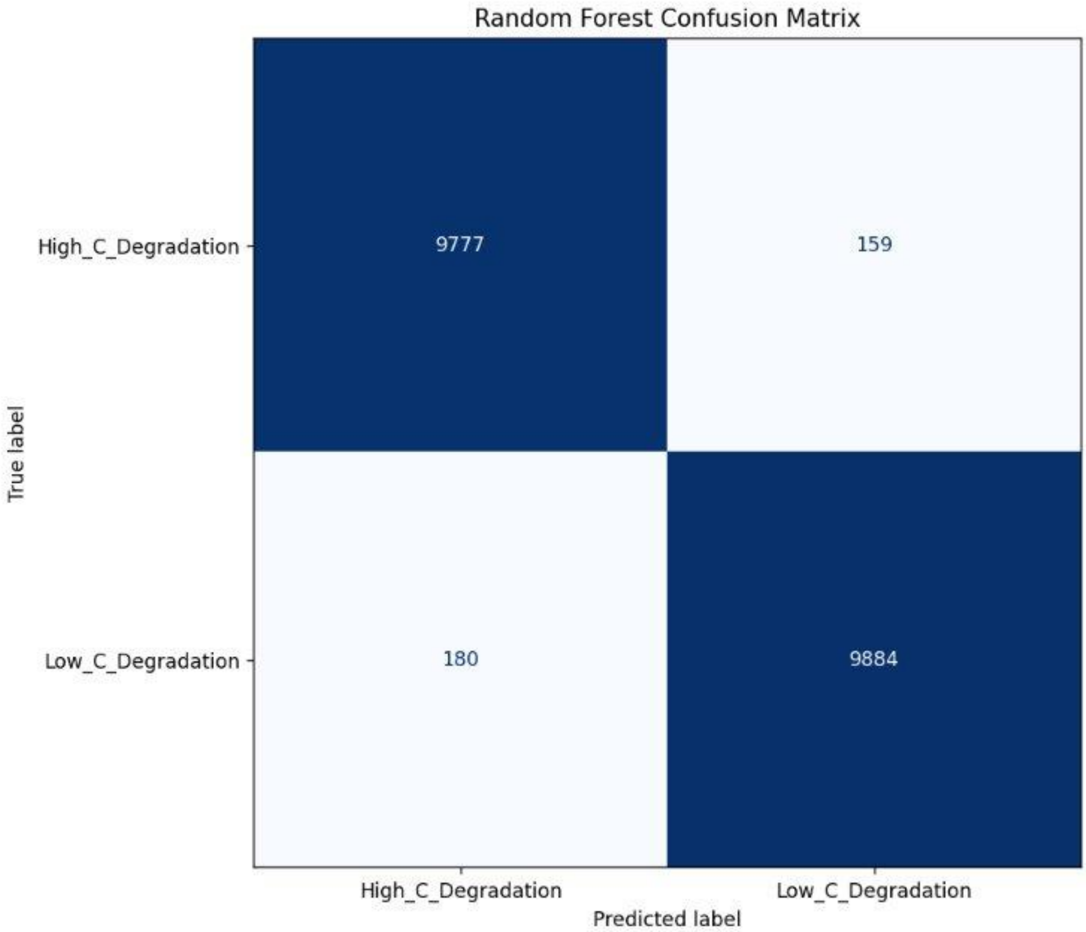
```
Model Performance on Test Set:
R² Score: 0.9996
MAE: 0.0820
RMSE: 0.1032

Train vs Test R²:
Train: 0.9996 | Test: 0.9996
```

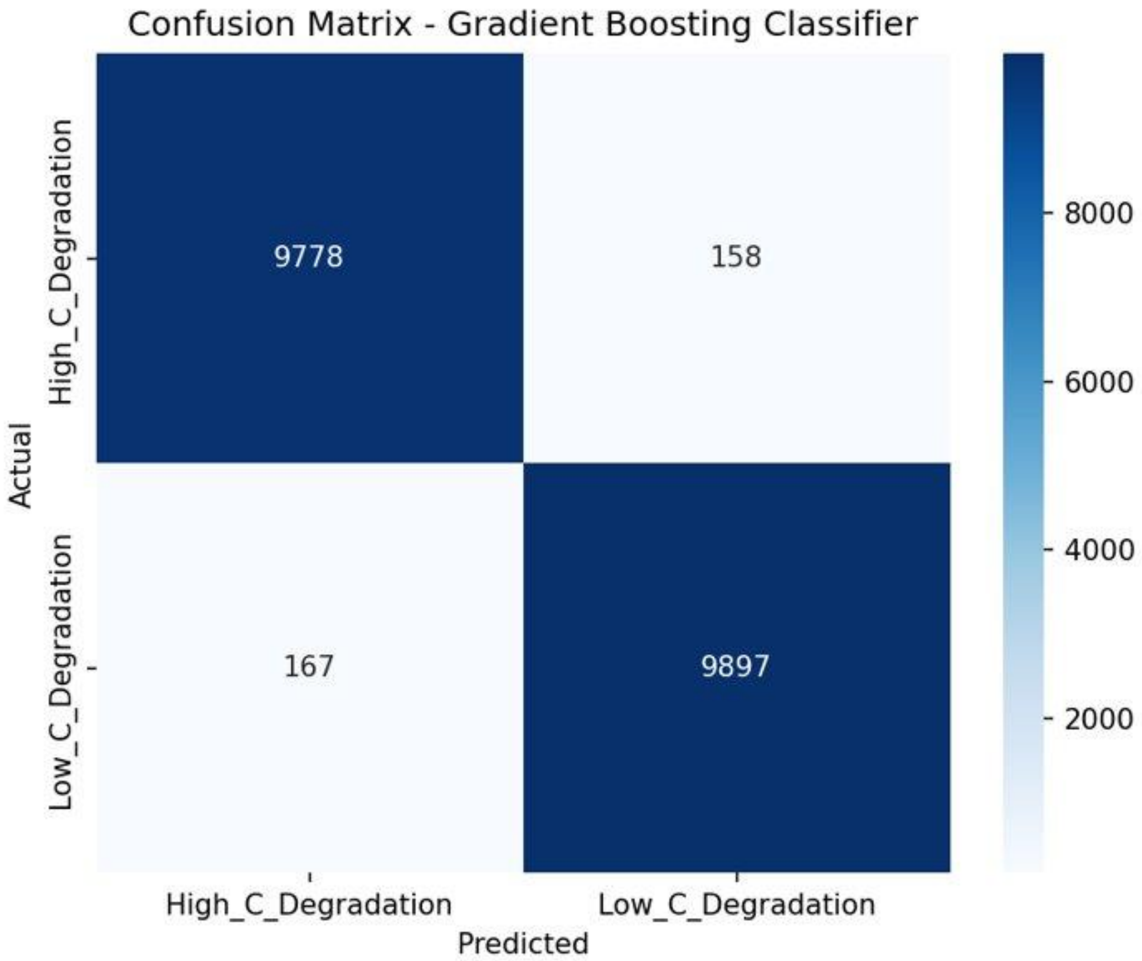


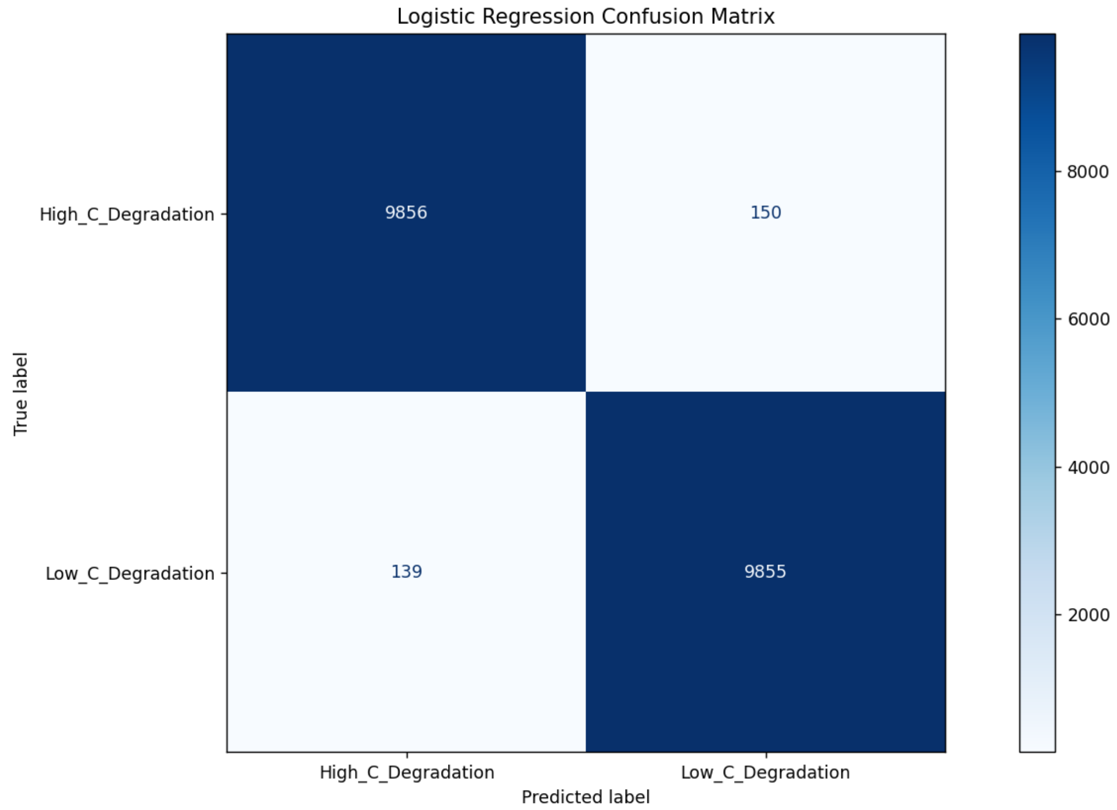
All tested models (Linear Regression, Random Forest, XGBoost) are highly accurate for this prediction task, with **R2 scores all above 0.9995,** indicating an almost perfect model fit.

CLASSIFICATION RESULTS:



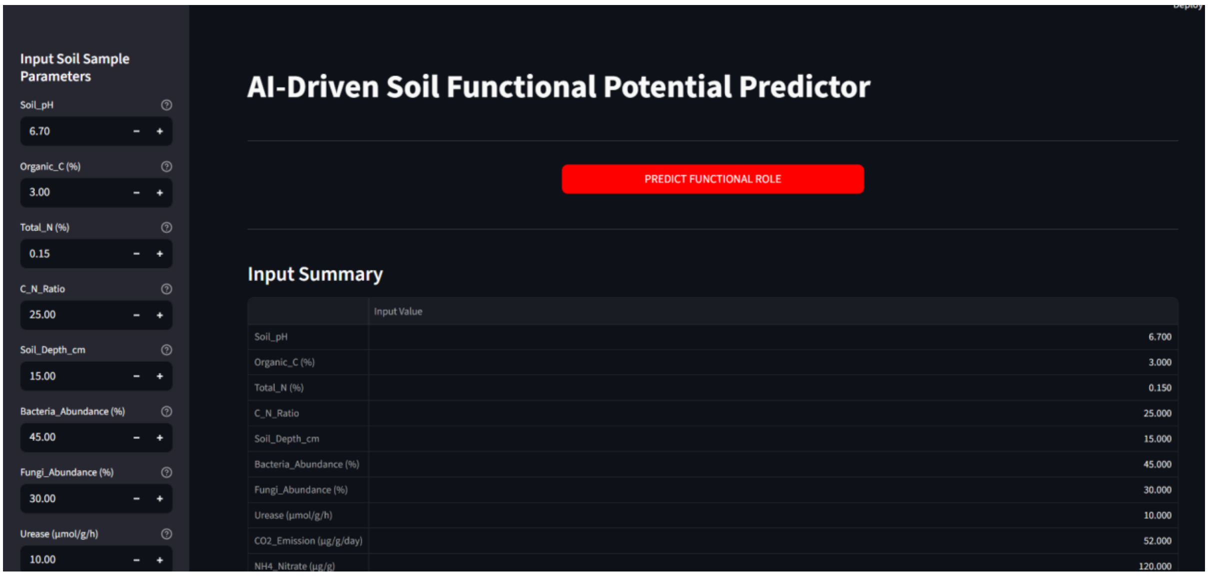
Random Forest Confusion Matrix
9777
159
180
9884
High_C_Degradation
Low_C_Degradation
True label
High_C_Degradation
Low_C_Degradation
Predicted label


Confusion Matrix - Gradient Boosting Classifier
Actual
High_C_Degradation
Low_C_Degradation
9778
158
167
9897
High_C_Degradation
Low_C_Degradation
Predicted
8000
6000
4000
2000



Among the trained models , Random Forest , Gradient Boosting , Logistic Regression

The Gradient Boosting model demonstrated **excellent predictive power**, achieving an overall **accuracy of 98.38%** on the test data. This indicates it is highly effective at classifying soil samples into 'High_C_Degradation' and 'Low_C_Degradation' roles based on their microbial and chemical properties.



A user-friendly Streamlit web application was also built to make these predictions interactive and accessible.

CONCLUSION :

This project develops high-accuracy machine learning models, including Linear Regression, Random Forest, and XGBoost, to predict soil CO_2 emissions , functional role prediction using soil microbe and chemical data. A user-friendly Streamlit web application was also built to make these predictions interactive and accessible. A significant next step would be to incorporate time-series data to model how functional roles change in response to environmental shifts, such as seasonal variations or different agricultural treatments. Furthermore, integrating other 'omics' data (e.g., metagenomics, metatranscriptomics, metaproteomics) would provide a more holistic understanding of the genetic potential and expressed functions of the microbes.

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